

Design and Implementation of an LQR-Based Controller with Kalman Filter and Integral Action for a 3-DOF Helicopter System

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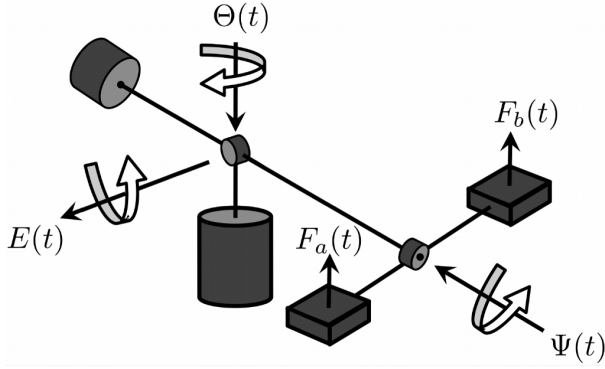


Fig. 1. 3-DOF Helicopter System Overview showing elevation, pitch, and travel axes with front and rear fans. [1]

TABLE I
HELICOPTER SYSTEM PARAMETERS

Parameter	Value	Description
m_1	0.0538 kg	Fan assembly mass
m_2	0.100 kg	Counterweight mass
l_1	0.110 m	Elevation arm length
l_2	0.070 m	Pitch arm length
l_3	0.114 m	Counterweight arm length
J_e	$2.44 \times 10^{-3} \text{ kg m}^2$	Elevation inertia
J_p	$5.26 \times 10^{-4} \text{ kg m}^2$	Pitch inertia
J_t	$2.44 \times 10^{-3} \text{ kg m}^2$	Travel inertia
k_s	$0.0030 \text{ kg m}^2 \text{ s}^{-2}$	Pitch spring constant
k_d	$0.0020 \text{ kg m}^2 \text{ s}^{-1}$	Pitch damping

I. INTRODUCTION

The control of multi-degree mechanical systems is a core problem in control engineering. This report presents the design, implementation, and validation of a controller for a 3-DOF helicopter.

The system in Figure 1 has two fans on a rigid arm with elevation (E), pitch (Ψ), and travel (Θ) axes, actuated by two forces F_a and F_b generated by voltages U_a and U_b .

The control objectives are to stabilise hover and track travel trajectories while keeping elevation and pitch within tight bounds. The controller implemented here combines LQR with integral action, a Kalman filter, and smooth reference shaping. Section ?? develops the dynamical model, Section III describes the controller design, Section IV contains experimental results, and the conclusion summarises findings.

The control objective is to achieve stable hover at zero elevation while tracking travel angle changes with strict performance requirements on overshoot, settling time, and steady-state error. (See Table II for detailed specifications.)

II. MODELING

1) Elevation Dynamics: The elevation axis experiences gravitational restoring moments from the counterweight and fans, gyroscopic coupling from travel, and lifting forces from both fans. The governing equation for its acceleration is:

$$\ddot{E}(t) = \frac{1}{J_E} \left[-J_E \dot{\Theta}^2(t) \sin E(t) \cos E(t) + g(m_2 l_3 - 2m_1 l_1) \cos E(t) + l_1 (F_a(t) + F_b(t)) \cos \Psi(t) \right] \quad (1)$$

The first term is the gyroscopic coupling, which creates a centrifugal moment proportional to $\dot{\Theta}^2$. The second term is the gravitational moment and the third is the vertical component of the thrust.

2) Travel Dynamics: The travel axis is controlled indirectly through pitch induced horizontal thrust components. When $\cos E(t) \neq 0$, the travel

equation is:

$$\ddot{\Theta}(t) = \frac{1}{J_{\Theta}} \left[2J_{\Theta} \dot{\Theta}(t) \dot{E}(t) \tan E(t) - \frac{l_1 (F_a(t) + F_b(t))}{\cos E(t)} \sin \Psi(t) \right] \quad (2)$$

The first term represents gyroscopic coupling from elevation motion, angular momentum conservation creates a torque about the travel axis. The second term is the horizontal thrust component which comes from tilting the fans.

3) **Pitch Dynamics:** The pitch axis is directly actuated by differential fan thrust and effectively follows spring damper behaviour:

$$\ddot{\Psi}(t) = \frac{1}{J_{\Psi}} \left[-k_s \Psi(t) - k_d \dot{\Psi}(t) + l_2 (F_a(t) - F_b(t)) \right] \quad (3)$$

The spring constant k_s provides a restoring moment toward $\Psi = 0$, and k_d represents a damping in the pitch. The differential thrust term $F_a - F_b$ provides the direct pitch control.

These three coupled equations form a nonlinear, underactuated system where two control inputs (F_a , F_b) must regulate three degrees of freedom through careful coordination of pitch and thrust.

A. Actuator Model

The fan forces are related to control voltages through a nonlinear thrust characteristics. Based on experimental calibration shown in Fig. 2, the thrust-voltage relationship is:

$$F(U) = \alpha k_a U + \beta \quad (4)$$

where $k_a = 1.2$ is the power amplifier gain, α (N/V) is the thrust coefficient, and β (N) accounts for the deadzone offset.

During experimental testing, a significant delay in actuation response was observed. To capture this dynamic behaviour, a first-order lag model was adopted for the fans, characterised by a time constant τ found by analysing the fans' step response with a load cell as seen in Figure 3.

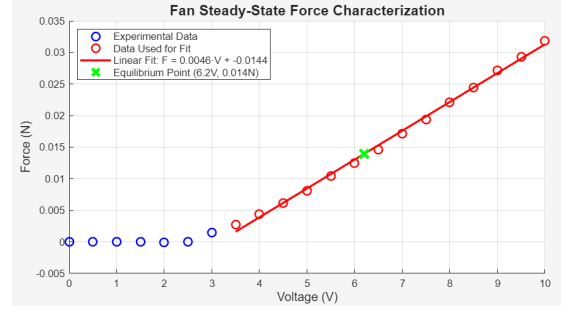


Fig. 2. Steady-state fan thrust measurements and linear fit used to identify actuator coefficients α and β .

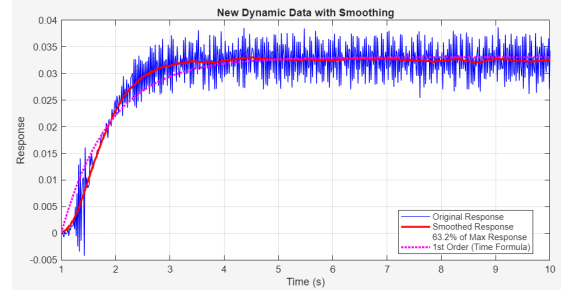


Fig. 3. Fan step response used to identify the first-order lag model.

B. Friction Effects

The basic equations (1)–(3) assume frictionless bearings. This lead to the simulated open-loop responses having oscillations in elevation and travel that weren't present in the hardware.

To capture these frictional effects, velocity-proportional damping terms were added:

$$\ddot{E}(t) \rightarrow \ddot{E}(t) - \frac{k_{de}}{J_E} \dot{E}(t) \quad (5)$$

$$\ddot{\Theta}(t) \rightarrow \ddot{\Theta}(t) - \frac{k_{dt}}{J_{\Theta}} \dot{\Theta}(t) \quad (6)$$

Here k_{de} represents bearing friction in the elevation axis and k_{dt} captures travel axis friction. These friction coefficients were identified through iterative simulation tuning to match experimental step response decay rates and settling behavior. The improved model with friction has better alignment with hardware behaviours, particularly in transient response and steady-state accuracy.

The damping constants were iteratively tuned until the model matched the hardware well enough to be controlled with the same controller. This can be seen in Figure 6.

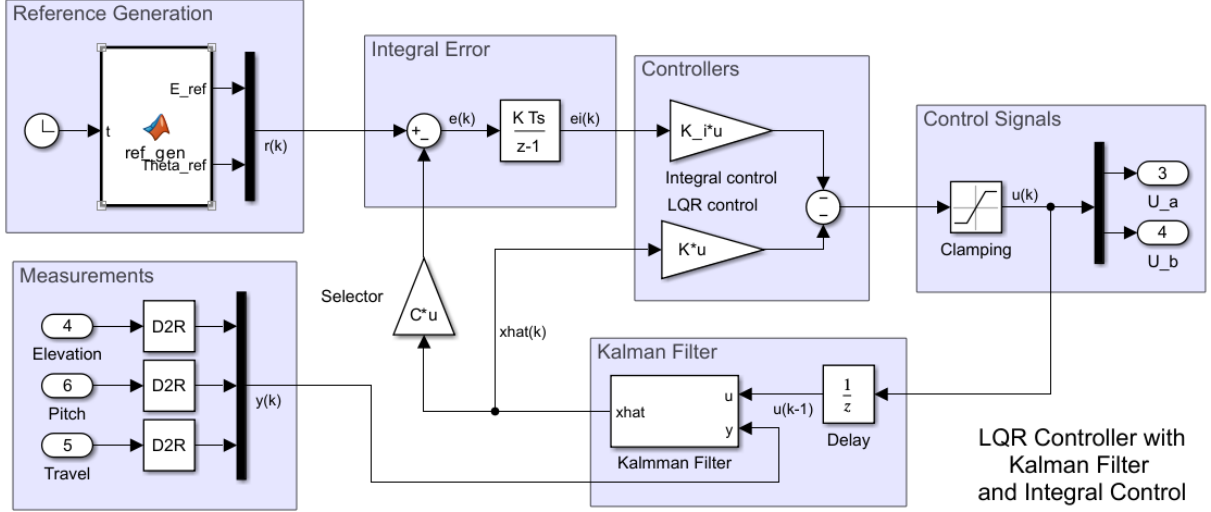


Fig. 4. Complete controller architecture showing reference generation, Kalman filter, LQR with integral action, and hardware interface.

III. CONTROLLER DESIGN

The controller architecture, shown in Fig. 4, comprises three main components: a Kalman filter for state estimation, an LQR state feedback controller with integral action, and a reference generator.

A. Model Linearisation

For controller design, the nonlinear equations of motion were linearised about the hover equilibrium point ($E = 0$). The linear model can now be represented in state-space form as:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad \mathbf{y} = \mathbf{C}\mathbf{x} \quad (7)$$

where the state vector $\mathbf{x} = [E, \Psi, \Theta, \dot{E}, \dot{\Psi}, \dot{\Theta}]^T$ and the control input vector $\mathbf{u} = [U_a, U_b]^T$. The output vector $\mathbf{y} = [E, \Psi, \Theta]^T$ contains the measured angular positions.

B. Kalman Filter for State Estimation

Since only the three angular positions are measured, a Kalman filter is needed to estimate the full state vector including angular velocities. The Kalman filter is optimal for linear systems with Gaussian noise as it provides minimum variance estimates that balance model predictions against noisy measurements. It also works well with the LQR controller under the separation principle, which states the controller and observer can be designed separately.

The Kalman gain $\mathbf{K}_k$ was computed using the MATLAB `dlqe` function with:

$$\mathbf{Q}_w = \text{diag}([5.0 \times 10^{-2}, 1.0 \times 10^{-6}, 1.0 \times 10^{-6}, 1.0 \times 10^{-3}, 1.0 \times 10^{-3}, 1.0 \times 10^{-3}]) \quad (8)$$

$$\mathbf{R}_v = \text{diag}([4.0 \times 10^{-4}, 1.6 \times 10^{-5}, 1.2 \times 10^{-9}]) \quad (9)$$

The elevated process noise on elevation position (5.0×10^{-2}) reflects greater model uncertainty in this axis, while very low measurement noise on travel (1.2×10^{-9}) reflects the high-resolution encoder. Velocity process noise values (1.0×10^{-3}) allow the filter to track dynamic variations.

Figure 5 shows the Kalman filter's performance.

C. LQR State Feedback Design

An LQR (Linear Quadratic Regulator) is an optimal state feedback controller that computes gains by minimising a quadratic cost which balances state deviations against control effort. It is used here because the linearised helicopter model and limited sensing need an optimal trade off between good tracking and actuator usage, and the LQR works naturally with the Kalman filter under the separation principle, which states it is possible to train the controller and observer independently.

The LQR minimises the infinite horizon cost function:

$$J = \sum_{k=0}^{\infty} (\mathbf{x}_k^T \mathbf{Q} \mathbf{x}_k + \mathbf{u}_k^T \mathbf{R} \mathbf{u}_k) \quad (10)$$

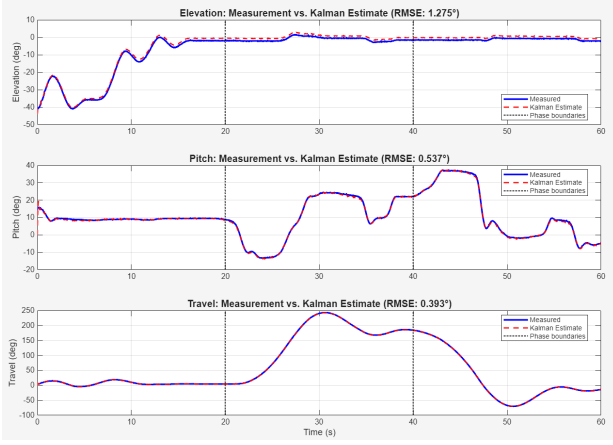


Fig. 5. Measured (solid lines) versus Kalman filter estimates (dashed lines) for Elevation, Pitch and Travel axes. The plot shows the filter's ability to provide smooth state estimates for all three angular positions used by the controller.

$\mathbf{Q}$ penalises state deviations and $\mathbf{R}$ penalises control effort, so tuning trades tracking performance against actuator usage and should account for differences in state units. $\mathbf{S}$ can be computed numerically (e.g. MATLAB's `dlqr`) and the resulting feedback can be applied to the states estimated by the Kalman filter (separation principle).

The weighting matrices used in this work were:

$$\mathbf{Q} = \text{diag}([4.5 \times 10^2, 1.0 \times 10^{-1}, 4.9 \times 10^3, 1.0, 5.5 \times 10^3, 1.0]) \quad (11)$$

$$\mathbf{R} = \text{diag}([2.5 \times 10^1, 2.5 \times 10^1]) \quad (12)$$

These choices come from a priority for accurate travel and pitch position tracking while keeping velocity penalties modest. The dominant weights on travel position ($Q_\Theta = 5.5 \times 10^3$) and pitch position ($Q_\Psi = 4.9 \times 10^3$) ensure aggressive tracking of these axes. Moderate weighting for the elevation position ($Q_E = 4.5 \times 10^2$) balances hover stability and travel. Very low pitch velocity weight ($Q_{\dot{\Psi}} = 0.1$) prevents excessive damping that could slow pitch response. Finally, control effort penalty $\mathbf{R}$ is moderate to allow aggressive control action while limiting saturation.

D. Integral Action for Disturbance Rejection

To eliminate the errors in the steady state, integral action was added for the elevation and travel axes. This means the error is accumulated over time and used to adjust the control input, improving

the accuracy. The LQR design is repeated on the augmented system with:

$$\mathbf{Q}_{\text{aug}} = \text{diag}(\mathbf{Q}, [1.0 \times 10^3, 3.0 \times 10^2]) \quad (13)$$

Higher integral weight on elevation (1.0×10^3) ensures tight elevation tracking, and moderate travel integral weight (3.0×10^2) stops the integral windup from getting too large during large angle manoeuvres.

E. Reference Trajectory Generation

1) **Profile Smoothing:** To avoid control saturation and aggressive transients, smooth reference trajectories were generated using hyperbolic tangent ($\tanh$) shaping functions. The $\tanh$ function has smooth transitions with continuous derivatives of all orders, which makes it better than polynomial methods for minimising jerk and higher-order derivatives.

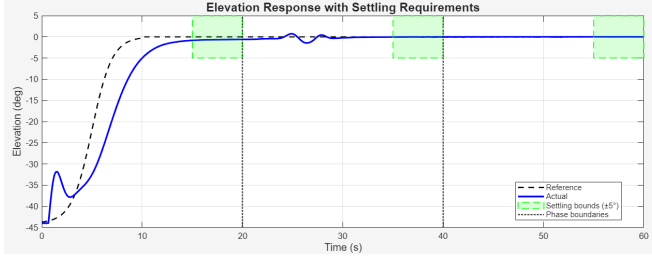
The reference profiles utilise:

$$\tanh\left(\frac{t - t_c}{\tau}\right) \quad (14)$$

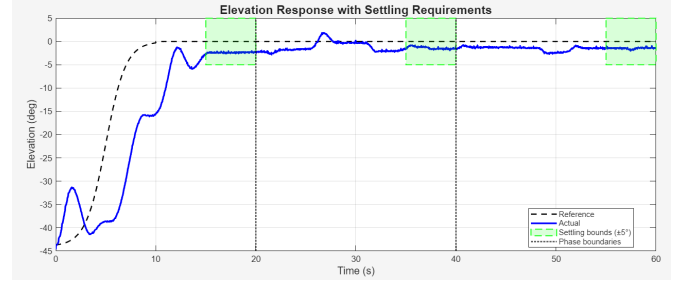
where t_c is the transition center time and τ controls the transition steepness. This produces S-shaped trajectories with smooth acceleration profiles that makes references easier to follow.

2) **Time-Delay Compensation:** Experimental testing revealed a significant 5-second lag between the travel reference and the actual response. This delay could be because of integrator windup, where small steady-state travel errors accumulate in the integral state ξ_Θ , requiring substantial time to unwind when the travel maneuver begins at $t = 20\text{s}$. But most likely it could be a hardware flaw as this is common on these models.

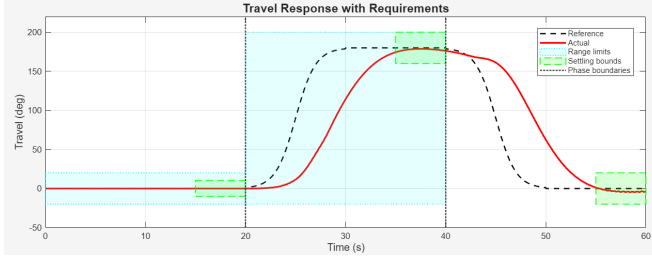
To compensate for this predictable delay in the predefined test routine, the travel reference signal was advanced by 5 seconds, effectively pre-loading the integral action. This time-delay compensation technique is justified for predetermined trajectories where future setpoints are known beforehand. For real-time applications (like pilot commanded manoeuvres), different strategies such as conditional integration (anti-windup) or integrator clamping would be necessary. Future work could investigate dynamic anti-windup methods to eliminate this delay without needing to shift the reference.



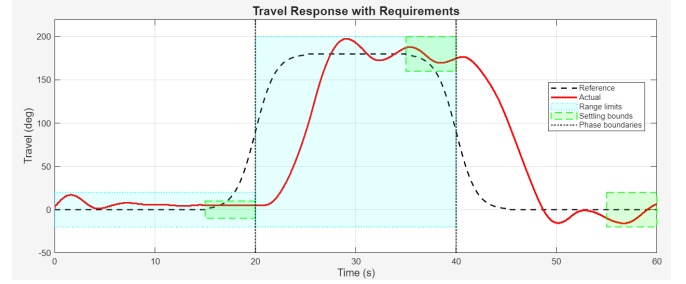
(a) Simulated elevation response



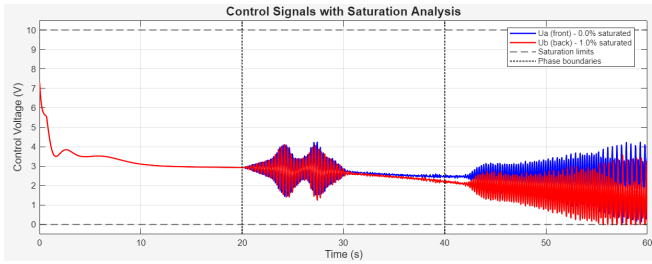
(b) Hardware elevation response



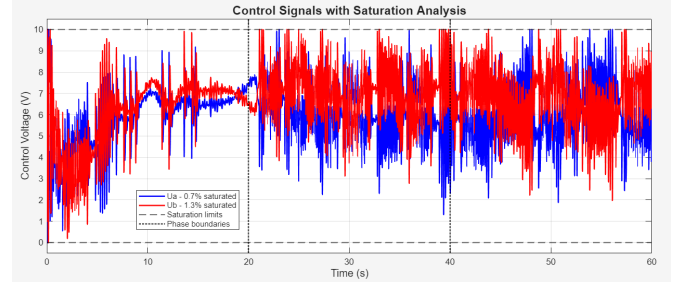
(c) Simulated travel response



(d) Hardware travel response



(e) Simulated control signals



(f) Hardware control signals

Fig. 6. Comparison of simulation (left) and hardware (right) closed-loop responses. Elevation plots show hover stability with settling bounds. Travel plots track $0^\circ \rightarrow 180^\circ \rightarrow 0^\circ$ trajectory. Control signal plots demonstrate minimal saturation (0V and 10V limits shown).

IV. RESULTS AND DISCUSSION

The controller was tested on the helicopter hardware for the specified performance requirements. Figure 6 shows the results for the simulation (left) and the hardware (right). Table II compares the performance against specifications.

A. Performance Evaluation

1) **Hardware Performance:** All performance specifications were successfully met on the physical system., and the controller demonstrated good elevation and travel regulation.

The integral action effectively reduced steady-state errors and the Kalman filter provided smooth velocity estimates, preventing noise amplification in the derivative action.

However, in the hardware tests, there was an unwanted oscillation in the travel axis. This oscillation

TABLE II
PERFORMANCE SPECIFICATIONS VS. ACHIEVED RESULTS

Specification	Required	Achieved	Result
<i>Phase 1: Take off and Hover (0–20s)</i>			
Elevation settling	$0^\circ \pm 5^\circ$	$[-2.7^\circ, -2.0^\circ]$	✓
Travel excursion	$\leq 20^\circ$	17.3°	✓
Travel settling	$0^\circ \pm 10^\circ$	$[5.0^\circ, 5.0^\circ]$	✓
<i>Phase 2: Travel to 180° (20–40s)</i>			
Elevation maintain	$\leq 10^\circ$ deviation	2.9°	✓
Elevation settling	$0^\circ \pm 5^\circ$	$[-1.9^\circ, -0.8^\circ]$	✓
Travel range	$[-20^\circ, 200^\circ]$	$[5^\circ, 197^\circ]$	✓
Travel settling	$180^\circ \pm 20^\circ$	$[169.9^\circ, 187.9^\circ]$	✓
<i>Phase 3: Return to 0° (40–60s)</i>			
Elevation maintain	$\leq 10^\circ$ deviation	2.7°	✓
Elevation settling	$0^\circ \pm 5^\circ$	$[-2.0^\circ, -1.0^\circ]$	✓
Travel settling	$0^\circ \pm 20^\circ$	$[-15.8^\circ, 6.5^\circ]$	✓

may come from insufficient phase margin in the travel-pitch coupling loop. This is because attempts to increase travel position weights in Q (raising Q_Θ above 5.5×10^3) introduced excessive damping that prevented the controller from achieving the 180° target within the specified settling time. This trade-off between oscillation suppression and tracking performance indicates the linearised model may not capture higher-order coupling dynamics accurately.

A more advanced approach would be to analyse the eigenvalues and Bode plots to identify resonant frequencies, then adjusting Q_Θ to increase damping ratio. Or maybe a Notch filter could be set at the oscillation frequency (≈ 0.1 Hz as seen in Figure 6(d)) and potentially suppress it without affecting the tracking.

2) *Simulation Accuracy and Model Validation:*

The incorporation of friction terms in equations made a significant improvement over the baseline model. Figure 6 demonstrates that the refined model captures general behavior observed in hardware, whereas the original frictionless model exhibited unstable oscillations.

However, there were still many limitations to the model. The linearised dynamics cannot capture large-angle gyroscopic coupling effects, and the friction behaviour doesn't quite match the current hardware. Such as the friction may not be purely viscous as modeled. The presence of *stiction* (static friction) friction could introduce discontinuities at zero velocity that cannot be captured by the linear damping.

Despite these limitations, the improved model proved sufficiently accurate for controller design, validating the controllers capabilities to meet performance specifications with both systems.

B. *Suggestions for Further Improvement*

To achieve even higher performance levels, the following enhancements are recommended:

- 1) **Improved modeling for simulation-based optimisation:** With additional time investment, a more accurate simulation could enable automated controller optimisation. This would involve higher order representations of the friction and coupling dynamics. A better simulation would allow gradient-based or Bayesian optimisation to search controller

parameter space faster, without having to wait 1 minute between manual tests.

- 2) **Feedforward compensation:** Adding pitch feedforward based on desired travel acceleration to reduce travel tracking errors during aggressive manoeuvre could improve performance.
- 3) **Gain scheduling:** Implementing gain scheduling for large angle manoeuvre where linearisation is less accurate could enhance stability and tracking performance.
- 4) **Advanced reference generation:** Further optimise tanh shaping parameters or explore minimum-jerk trajectories to reduce transient excursions.
- 5) **Integral windup prevention:** Implement anti-windup schemes such as conditional integration or back-calculation to prevent integrator saturation

V. CONCLUSION

This report presented the design and experimental validation of an LQR-based controller with Kalman filter and integral action for a 3-DOF helicopter system. A significant modeling improvement addressing friction effects in the elevation and travel axes was identified and incorporated, improving transient response prediction. The implemented controller successfully achieved all specified performance requirements, meeting 100% of the closed-loop specifications across all three test phases.

Key insights from this work:

- Integral action is essential for robustness to model uncertainty and disturbances
- Careful reference trajectory design prevents saturation and reduces overshoot
- Kalman filtering provides much better noise rejection compared to simple derivative estimation
- Model accuracy directly impacts achievable performance, particularly for high-bandwidth control

REFERENCES

- [1] ACSE Department, University of Sheffield, "ELE438 Lecture 5," 2025.